

PETROPAVL, Kazakhstan

WMO: 286760

Lat: 54.7923N

Lon: 69.1174E

Elev: 101

StdP: 100.12

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.4	(c) -30.3	(d) -36.3	(e) 0.1	(f) -33.2	(g) -33.3	(h) 0.2	(i) -30.1	(j) 12.3	(k) -9.8	(l) 11.1	(m) -10.4	(n) 2.3	(o) 90	(p) 0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.4	(c) 30.8	(d) 19.8	(e) 28.9	(f) 19.2	(g) 27.2	(h) 18.4	(i) 21.4	(j) 27.9	(k) 20.5	(l) 26.7	(m) 19.6	(n) 25.4	(o) 4.0	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.1	24.4	18.2	13.3	23.5	17.3	12.5	22.7	62.8	28.1	59.5	26.7	56.4	25.5	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.3	9.5	8.4	-36.8	33.9	3.3	2.0	-39.2	35.3	-41.2	36.4	-43.0	37.5	-45.5	39.0
			-36.8	23.2	3.3	1.3	-39.1	24.2	-41.0	24.9	-42.9	25.7	-45.2	26.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-17.3	-15.2	-6.9	4.5	12.9	18.0	19.3	17.4	11.3	4.1	-6.5	-13.6
	DBStd	14.38	7.46	7.21	5.87	6.14	5.07	4.38	3.55	4.18	4.65	5.08	7.70	7.44
	HDD10.0	3751	846	706	523	184	29	2	0	1	38	193	495	733
	HDD18.3	5974	1104	939	782	415	179	58	33	70	216	442	745	992
	CDD10.0	986	0	0	0	19	119	243	288	230	77	9	0	0
	CDD18.3	167	0	0	0	0	11	49	62	41	5	0	0	0
	CDH23.3	1686	0	0	0	7	177	518	548	372	63	1	0	0
	CDH26.7	491	0	0	0	1	42	174	151	107	15	0	0	0
Wind	WSAvg	4.1	3.9	4.1	4.6	4.6	4.6	3.9	3.5	3.5	3.8	4.1	4.1	4.1
Precipitation	PrecAvg	368	20	15	15	19	31	42	62	50	29	31	28	21
	PrecMax	594	42	47	41	45	75	111	208	121	76	97	68	57
	PrecMin	246	1	0	0	0	0	7	10	2	4	2	0	0
	PrecStd	80	11	9	11	12	19	24	38	26	17	20	14	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.4	1.2	7.5	23.8	30.5	32.9	32.9	31.9	28.8	20.5	8.1	1.2
		MCWB	0.0	0.0	4.3	13.0	17.5	20.1	21.2	19.9	16.2	11.6	6.1	0.3
	2%	DB	-2.0	-0.5	3.6	20.0	27.5	30.9	30.0	29.6	24.9	17.2	5.7	0.1
		MCWB	-2.5	-1.3	1.7	11.3	15.7	19.3	20.5	19.8	14.9	10.5	4.3	-0.6
	5%	DB	-3.7	-2.3	2.3	16.9	25.2	28.7	28.1	27.4	21.7	14.2	4.0	-1.3
		MCWB	-4.3	-3.0	0.9	10.1	15.0	19.2	19.9	19.1	14.1	8.9	2.8	-2.0
	10%	DB	-5.9	-4.4	1.1	13.8	22.6	26.2	26.5	25.1	19.0	11.5	2.2	-3.7
		MCWB	-6.4	-5.1	0.0	8.7	13.6	18.1	19.2	18.0	12.8	7.8	1.1	-4.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.1	0.2	4.3	14.2	18.9	22.2	22.8	22.2	17.8	13.0	6.2	0.6
		MCDB	0.3	1.1	7.0	21.2	27.4	29.7	28.9	29.2	25.0	18.4	7.5	0.9
	2%	WB	-2.5	-1.3	2.0	12.1	17.1	21.0	21.6	20.9	15.9	11.2	4.4	-0.6
		MCDB	-2.1	-0.7	3.1	18.4	25.3	28.1	27.7	27.0	22.6	15.8	5.4	0.0
	5%	WB	-4.3	-3.0	1.0	10.5	15.7	19.7	20.7	19.8	14.6	9.4	2.9	-2.1
		MCDB	-3.7	-2.4	2.1	16.3	23.3	26.6	26.6	25.6	20.5	13.4	3.9	-1.3
	10%	WB	-6.4	-5.1	0.0	8.8	14.4	18.6	19.8	18.7	13.3	8.0	1.2	-4.3
		MCDB	-6.0	-4.4	1.0	13.6	21.2	24.9	25.3	24.0	18.2	11.0	2.0	-3.7
Mean Daily Temperature Range	5% DB	MDBR	7.8	9.0	8.9	9.7	11.9	11.2	10.4	10.8	10.3	8.0	6.5	7.5
		MCDBR	8.0	8.1	7.6	14.4	15.9	14.4	13.1	14.1	14.7	13.2	6.6	6.9
		MCWBR	7.9	7.7	6.2	7.5	7.4	6.4	5.9	6.3	7.2	7.8	5.6	6.7
	5% WB	MCDBR	7.9	7.9	7.0	13.4	14.5	12.7	11.7	12.4	13.4	12.1	6.2	6.6
		MCWBR	7.8	7.6	6.0	7.5	7.6	6.4	5.8	6.1	7.2	7.5	5.5	6.5
Clear-Sky Solar Irradiance	taub		0.286	0.320	0.376	0.415	0.409	0.411	0.433	0.437	0.381	0.359	0.323	0.289
	taud		2.047	2.010	1.942	2.103	2.221	2.290	2.219	2.178	2.293	2.226	2.109	2.019
	Ebn at Noon		612	727	766	802	834	837	807	774	774	686	559	505
	Edh at Noon		69	103	140	138	130	123	130	128	100	84	66	57
All-Sky Solar Radiation	RadAvg		0.99	2.07	3.61	4.60	5.59	6.00	5.60	4.54	3.14	1.61	0.79	0.63
	RadStd		0.07	0.10	0.29	0.48	0.43	0.46	0.41	0.41	0.25	0.25	0.11	0.09

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page